

Model-free forecasting of partially observable spatiotemporally chaotic systems

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ABSTRACT

Reservoir computing is a powerful tool for forecasting turbulence because its simple architecture has the computational efficiency to handle high-dimensional systems. Its implementation, however, often requires full state-vector measurements and knowledge of the system nonlinearities. We use nonlinear projector functions to expand the system measurements to a high dimensional space and then feed them to a reservoir to obtain forecasts. We demonstrate the application of such reservoir computing networks on spatiotemporally chaotic systems, which model several features of turbulence. We show that using radial basis functions as nonlinear projectors enables complex system nonlinearities to be captured robustly even with only partial observations and without knowing the governing equations. Finally, we show that when measurements are sparse or incomplete and noisy, such that even the governing equations become inaccurate, our networks can still produce reasonably accurate forecasts, thus paving the way towards model-free forecasting of practical turbulent systems.

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1. Introduction

The availability of large datasets and advances in computational hardware have led to revolutionary applications of machine learning (ML), such as in speech recognition (Hinton et al., 2012) and medical diagnosis (Alber et al., 2019). This has led to renewed interest in the application of ML to predict the behaviour of turbulent flow systems, such as weather and climate forecasting (Kashinath et al., 2021; Pathak et al., 2022; Reichstein et al., 2019; Shi et al., 2017; Wan & Sapsis, 2017; Wan, Vlachas, Koumoutsakos, & Sapsis, 2018; Watt-Meyer et al., 2021). Turbulent systems are defined by processes spanning a wide range of spatial and temporal scales. These processes are coupled nonlinearly such that even small changes at the microscopic scale can lead to large changes at the macroscopic scale (Leith & Kraichnan, 1972), a phenomenon poetically termed the “butterfly effect”

by Lorenz (1993). Modelling such nonlinearly coupled dynamical systems with recurrent neural networks (RNNs) would require the network dimension to be exceedingly large, limiting their applicability to either coarse predictions (Shi et al., 2017; Wan & Sapsis, 2017) or necessitating dimensionality reduction of the input states (Chandramouli et al., 2020; Vlachas, Arampatzis, Uhler, & Koumoutsakos, 2022; Vlachas, Byeon, Wan, Sapsis, & Koumoutsakos, 2018).

Reservoir-computing-based RNNs (RC-RNNs) have been shown to be more computationally efficient at learning large chaotic systems as compared to backpropagation RNNs (Chattopadhyay, Hassanzadeh, & Subramanian, 2020; Vlachas et al., 2020). Their implementation, however, often requires knowledge of the type of nonlinearities in the governing equations (Chattopadhyay et al., 2020; Pathak, Hunt, Girvan, Lu, & Ott, 2018). This hinders the application of RC-RNNs to natural and engineering systems where full state-vector measurements are often not available and/or the governing equations are not known. In this paper, we introduce a modification to RC-RNNs whereby nonlinear projectors are used to first expand the observables to a high dimensional space before they are fed to the reservoir. We demonstrate the application of such networks for forecasting spatiotemporally chaotic systems,

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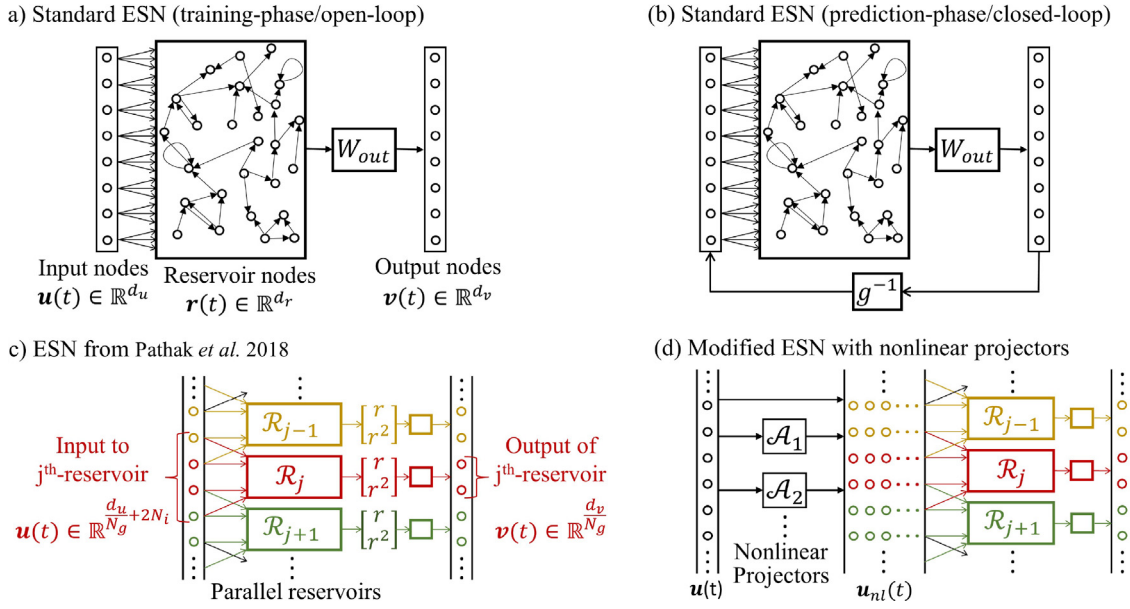


Fig. 1. (a) Schematic of a standard ESN. The input nodes are connected to a high-dimensional reservoir, which is the only hidden layer in the network. A large enough reservoir can capture the system nonlinearities from which the output (future predictions here) is obtained via linear readout. (b) The network in the prediction phase is closed-loop, i.e. the output of the network becomes its input at the next step. (c) Schematic of the ESN from Pathak et al. (2018); it has two improvements: (i) it is parallelized and (ii) the reservoir state is expanded to a higher-dimension (such as from r to $[r, r^2]$). (d) Schematic of the proposed modified ESN; the observed vector $\mathbf{u}(t)$ is expanded to $\mathbf{u}_{nl}(t)$ via nonlinear projectors $\mathcal{A}_1, \mathcal{A}_2$, etc (i.e. $\mathbf{u}_{nl}(t) = [u(t), \mathcal{A}_1(u(t)), \mathcal{A}_2(u(t)), \dots]$). This means that the nonlinearity in the modified network is introduced before the reservoir instead of after it as in the network in (c).

which are nonlinear dynamical systems with a large number of degrees of freedom that can model several important features of turbulence (Cross & Hohenberg, 1993; Yakhot, 1981). In particular, we explore the use of radial basis functions (RBFs), which has also inspired other RNNs for predicting chaotic systems (Itoh & Adachi, 2010; Tyshchenko, 2016; Zemouri, Racoceanu, & Zerhouni, 2003), as nonlinear projectors to enable the networks to capture complex system nonlinearities without knowing the governing equations. We then explore scenarios in which only sparse or noisy and incomplete measurements are available, and compare the predictions from RC-RNNs with those from conventional time-marching.

2. Network details

We choose the echo-state network (ESN) of Jaeger and Haas (2004) for its ability to forecast chaotic systems. However, we emphasize that our work is applicable to other RNNs as well. In ESNs the input state $\mathbf{u}(t) \in \mathbb{R}^{d_u}$, which consists of d_u concurrent measurements in space, is connected to a high-dimensional dynamical system, say $\mathcal{R}$, known as the reservoir. The reservoir has the activation state $\mathbf{r}(t) \in \mathbb{R}^{d_r}$, which is effectively a nonlinear function of the input history $\mathbf{u}(t), \mathbf{u}(t - \Delta t), \dots$, where Δt is a discrete time step (Eq. (1a)). The output $\mathbf{v}(t) \in \mathbb{R}^{d_v}$ is then obtained as a linear readout from $\mathbf{r}(t)$ using Eq. (1b).

$$\mathbf{r}(t) = \mathcal{N}(\mathbf{A}\mathbf{r}(t - \Delta t) + \mathbf{W}_{in}\mathbf{u}(t)), \quad (1a)$$

$$\mathbf{v}(t) = \mathbf{W}_{out}\mathbf{r}(t). \quad (1b)$$

Here, $\mathbf{W}_{in} \in \mathbb{R}^{d_r \times d_u}$ connects the input state $\mathbf{u}(t)$ to the reservoir nodes, $\mathbf{A} \in \mathbb{R}^{d_r \times d_r}$ is an adjacency matrix that interconnects the reservoir nodes, $\mathcal{N}$ is a known nonlinear function (tanh here) that acts on each element separately (i.e. $\mathcal{N}([a_1, a_2, \dots]) = [\mathcal{N}(a_1), \mathcal{N}(a_2), \dots]$) and $\mathbf{W}_{out} \in \mathbb{R}^{d_v \times d_r}$ connects the reservoir nodes to the output (see the schematic in Fig. 1(a)).

The elements of $\mathbf{W}_{in}$ and $\mathbf{A}$ are fixed by drawing them from random distributions, such as from a uniform distribution in

$[-1, 1]$. The elements of $\mathbf{A}$ are scaled to keep its spectral radius (ρ) usually below one so that the network satisfies the fading memory property, i.e. the future state depends only on a finite sequence of past inputs. Finally, the elements of $\mathbf{W}_{out}$ are optimized via a one-step linear regression that minimizes the mean-square error between $\mathbf{v}(t)$ and the target signal $\hat{\mathbf{v}}(t)$ during the training time $-T \leq t \leq 0$. In forecasting problems, which are of interest here, the target signal $\hat{\mathbf{v}}(t) = g(\mathbf{u}(t + 1))$, where g is a bijective function. Consequently, the network becomes closed-loop in the prediction phase, i.e. the model output becomes its input at the next step as shown in Fig. 1(b). See Appendix A for further details and Lukoševičius (2012) for practical guidance on designing ESNs.

If the complex nonlinearities of turbulent systems are to be captured, d_r may need to be several orders of magnitude larger than d_u . This can be computationally costly for turbulent systems for which d_u is already large. Recognizing this, Pathak et al. (2018) proposed two improvements (see Fig. 1(c)). First, they parallelized the ESNs based on the spatial locality of turbulent interactions, which facilitates parallel computing. The parallelized network has two additional hyperparameters: the number of parallel reservoirs N_g and the interaction length N_i . Second, they explicitly expanded the reservoir activation state to a higher-dimensional space such that the output is obtained as $\mathbf{v}(t) = \mathbf{W}_{out}[\mathbf{r}, \mathbf{r}^2]$. The main drawback of this approach is that in order to capture the system nonlinearities, the nonlinearity added explicitly should closely match that present in the system equations themselves (Chattopadhyay et al., 2020).

We will show that the network from Pathak et al. (2018) can be improved if the expansion to the high-dimensional space happens before the reservoir. In other words, we expand the observables (i.e. $\mathbf{u} \in \mathbb{R}^{d_u}$) to a high-dimensional space using nonlinear projector functions $\mathcal{A}_l$ ($l = 1, 2, \dots, L$). The expanded vector $\mathbf{u}_{nl} \in \mathbb{R}^{(L+1)d_u}$ is then the input to the reservoir from which the output is trained via linear readout (see Fig. 1(d)). The increase in the size of the input vector and the corresponding $\mathbf{W}_{in}$ matrix in the modified network do not add to the computational cost required to train the network; the computational cost is mostly dependent on the reservoir size and the length of the training time-series (see Appendix A for details).

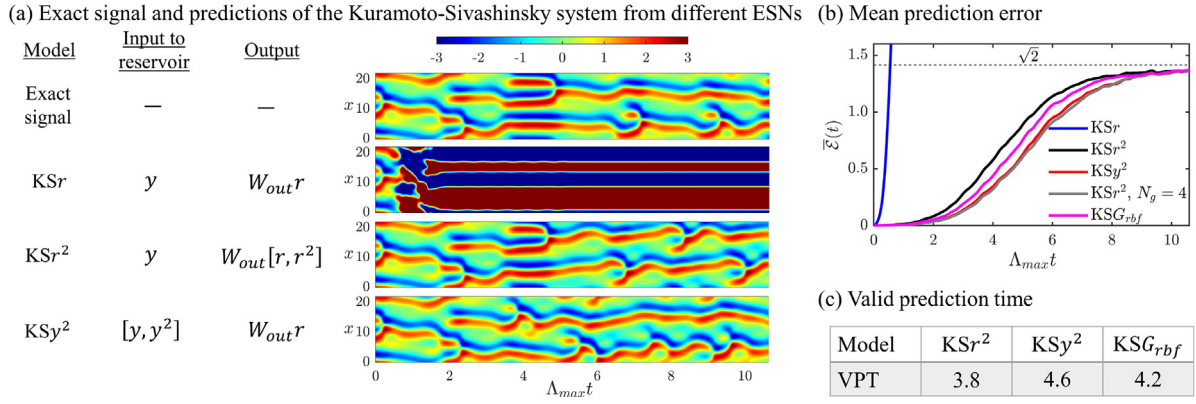


Fig. 2. (a) The exact signal for the KS system and its predictions from different RC-RNNs: KSR (an instance of a standard network shown in Fig. 1(a)), KSR² (an instance of the network from Pathak et al. (2018) shown in Fig. 1(c)) and KSY² (an instance of the proposed network shown in Fig. 1(d)). All networks here are of size $d_r = 1000$ and $N_g = 1$ (i.e. the reservoir size is only an order of magnitude larger than d_u). The KSR network, which has no explicitly added nonlinearity, cannot capture the system nonlinearities (unless maybe for much larger d_r). The networks with added square nonlinearities (KSR² and KSY²) are able to capture the system nonlinearities well. The mean prediction errors (b) and the valid prediction time (c) from the different networks are also shown. The modified network KSY² has relatively slower growth of error as compared to the network of Pathak et al. (2018) (KSR²). The improvement is significant considering that the KSR² network would require four times more computational power to achieve similar performance to the KSY² network. The KSG_{rbf} network is also an instance of the proposed network but with input to reservoir as $[y, \exp(-0.03y^2)]$, this network is discussed with Fig. 4 (d–f).

3. Predictions from existing and modified networks

We consider the Kuramoto–Sivashinsky (KS) equation as our first example:

$$y_t = -yy_x - y_{xx} - \nu y_{xxx}, \quad x \in [0, L]. \quad (2)$$

The scalar field $y(x, t)$ is periodic in space. The subscripts x and t denote the partial derivatives in space and time, respectively. The parameter ν acts like the viscosity in the Navier–Stokes equations (Bratanov, Jenko, Hatch, & Wilczek, 2013) and L is the system size with which the attractor dimension increases linearly (Manneville, 1985). We numerically solve Eq. (2) for $(\nu, L) = (1, 22)$ with $N = 128$ gridpoints and timestep $dt = 0.25$ (Kassam & Trefethen, 2005). In this study Δt of networks is set as dt and the time is nondimensionalized by the maximum Lyapunov exponent of the respective system in all the presented results.

Fig. 2(a) shows the exact signal from the simulations and its prediction from different ESNNs. All the networks feature a single reservoir of size $d_r = 1000$, which is only an order of magnitude larger than $d_u = 128$. The networks are trained for $-10000 \leq t \leq 0$, training with much smaller time history is possible but the predictive accuracy will then be lower as shown in Pathak et al. (2018). We note that the KSR network, which is an instance of the standard network (Fig. 1(a)), is unable to learn the system dynamics, i.e. the error in prediction increases rapidly and the corresponding forecast becomes physically uninterpretable. Nevertheless, the results drastically improve when the networks with added nonlinearities are used. Here KSR² and KSY² are instances of the network from Pathak et al. (2018) (Fig. 1(c)) and the proposed network (Fig. 1(d)), respectively. Both networks predict the exact signal in the short-term after which the patterns in the predictions appear to be spatiotemporally shifted. This shift is an inherent property of chaotic systems: their trajectories diverge exponentially in time even when they start from infinitesimally close initial conditions. The rate of separation is given by $\exp(\Lambda_{\max} t)$, where $\Lambda_{\max}$ is the maximal Lyapunov exponent and is positive for chaotic systems (Boffetta, Cencini, Falcioni, & Vulpiani, 2002).

We quantify the prediction error as $\mathcal{E}(t) = \sqrt{\langle (y(t) - \hat{y}(t))^2 \rangle} / \sigma_y$, where $\langle \cdot \rangle$ denotes spatial averaging and σ_y is the standard deviation of $y(x, t)$. To account for variability when comparing the performance of different ESNNs, we take the mean of the

error ($\bar{\mathcal{E}}(t)$) for a series of predictions (see Appendix B for details). Fig. 2(b) shows $\bar{\mathcal{E}}(t)$ for predictions from various ESNNs. The mean error increases exponentially from any infinitesimal differences in the predictions at $t = 0$. A slower increase in $\bar{\mathcal{E}}(t)$ would reduce the effect of both measurement and model uncertainties on future predictions. However, we note that the mean-square-error contains only limited information (Majda & Chen, 2018). For the predictions to be physically interpretable, their probability distribution should be close to that of the exact signal (DelSole, 2004). (This is the reason for the popularity of generative-adversarial networks (Goodfellow et al., 2014), which can give better interpretable results often at the cost of increased mean-square-error (Kim, Kim, Won, & Lee, 2021).) If the predictions are physically interpretable in the long-term, $\bar{\mathcal{E}}(t)$ would saturate to a value close to $\sqrt{2}$.

As expected from the predictions in Fig. 2(a), $\bar{\mathcal{E}}(t)$ from the standard network KSR not only increases rapidly but it also does not saturate to $\sqrt{2}$, i.e. the predictions are not interpretable. For all other networks used here, $\bar{\mathcal{E}}(t)$ saturates to $\sqrt{2}$ thus indicating statistically consistent predictions. We note that the modified network KSY² has smaller growth-rate of error as compared to that for the network from Pathak et al. (2018) (KSR²). In fact, the KSR² network requires almost four times more computational power (i.e. $N_g = 4$) to achieve a similar level of accuracy as the KSY² network. Although both networks have explicitly added square nonlinearities, the KSY² network outperforms the KSR² network. This is because in the KSY² network the reservoir receives more information about the system. In other words, the reservoir in the KSY² network is preconditioned with square nonlinearity and can thus better approximate the system nonlinearity.

Another metric recently introduced for evaluating the predictive performance of models is the valid prediction time (VPT) (Vlachas et al., 2020). It refers to the time at which $\bar{\mathcal{E}}(t)$ crosses a threshold; the threshold is set as 0.5 following Vlachas et al. (2020). The advantage of using VPT is that it is a scalar measure that quantifies the predictive accuracy of different methods. Networks with slower growth-rate of the mean-square error also have larger VPT as shown in Fig. 2(c) for the KS system; this trend is independent of the threshold value.

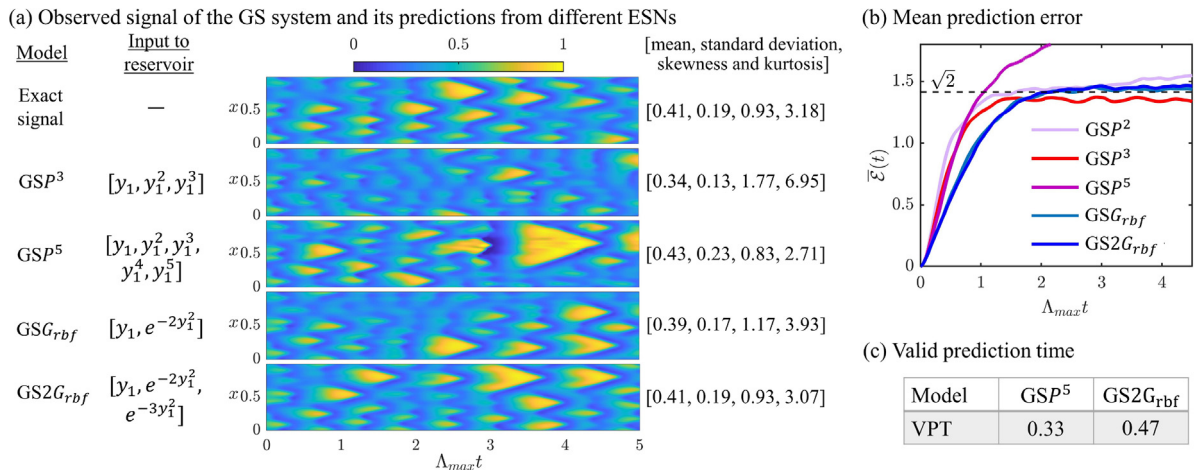


Fig. 3. (a) The observed signal, y_1 , of the GS system and its predictions from the GSP³, GSP⁵, GSG_{rbf} and GS2G_{rbf} networks of size $d_r = 1500$ and $N_g = 4$ (i.e. the total reservoir size is 6000, which is only an order of magnitude larger than d_u). Although all four networks can predict the short-term dynamics, the networks with monomial projectors (GSP³ and GSP⁵) fail to give statically consistent results while the networks with Gaussian RBF projectors (GSG_{rbf} and GS2G_{rbf}) give statistically consistent results. Also shown are the mean prediction errors (b) and the valid prediction time (c) from the different ESNs. In addition to statistically consistent results, the networks with Gaussian RBF projectors also show slower growth-rates of the error and hence higher VPTs.

4. Partially observable systems: selection of nonlinear projectors

For the KS system, we showed that the networks with explicitly added square nonlinearities (KSr² and KSy²) work well. As noted in Chattopadhyay et al. (2020), this is because the added nonlinearity in the networks matches the nonlinearity in the system equations (i.e. square nonlinearity in the KS system here). The form of system nonlinearity, however, is not always known; for example the governing equations are not known or full state-vector measurements are not available. To illustrate, we consider the Gray–Scott (GS) system, which models the reaction–diffusion of two species, as our second example,

$$\partial_t y_1 = (F - y_1 y_2^2) - F y_1 + D_1 \partial_{xx} y_1, \quad (3a)$$

$$\partial_t y_2 = y_1 y_2^2 - (F + k) y_2 + D_2 \partial_{xx} y_2, \quad (3b)$$

where $y_1(x, t)$ and $y_2(x, t)$ are the concentration of two chemical species. We choose $D_1 = 2.0e^{-5}$, $D_2 = 1.0e^{-5}$, $F = 0.015$, $k = 0.0415$ and $x \in [0, 1]$ with periodic boundary conditions for which the system exhibits annihilation type spatiotemporal chaos (Nishiura & Ueyama, 2001). We numerically solve the GS system with 200 Chebyshev points and $dt = 0.5$ and save the results on an equispaced grid containing 256 nodes (Driscoll, Hale, & Trefethen, 2014). The time evolution of the concentration of species one (y_1) is shown in Fig. 3(a).

We now assume that only y_1 measurements are available ($\mathbf{u}(t) = y_1(x, t)$), i.e. we do not know the full state-vector. One may argue that we can still form the full state-vector in terms of $(y_1, \partial_t y_1)$ or calculate y_2 using $(y_1, \partial_t y_1, \partial_{xx} y_1)$. This will, however, require knowledge of Eq. (3) and noise free measurements of y_1 with high spatiotemporal resolution. Neither are usually available in practical situations, nor are they needed by our method (see the results in Figs. 5 and 6).

With only y_1 measurements available, the form of system nonlinearity is not known. The challenge is therefore to select nonlinear projectors $\mathcal{A}_l$ that can enable the network to capture the nonlinear dynamics of the GS system from y_1 measurements alone. First we try monomial projectors, such as y_1^2, y_1^3 , etc., as used in new-generation-ESNs (Gauthier, Bollt, Griffith, & Barbosa, 2021) and often used to approximate nonlinearity in turbulent systems (Brunton, Proctor, Kutz, & Bialek, 2016). (GSPⁿ refers to an ESN with nonlinear projectors $\mathcal{A}_l = y_1^{l+1}$, $l = 1, 2, \dots, n - 1$.)

We note from Fig. 3(b) that the growth of $\bar{\epsilon}$ slows down slightly with increasing number of monomials but $\bar{\epsilon}$ does not saturate to $\sqrt{2}$ for any of the GSPⁿ networks. Fig. 3(a) shows poor quality of the predictions from the GSPⁿ networks. The statistics here is calculated based on the closed-loop predictions from $0 \leq t \leq 25000$. We find that adding more monomials or reservoir units does not significantly improve the quality of predictions from the GSPⁿ networks (results not shown).

Next, we try Gaussian RBFs as nonlinear projectors. This choice is inspired by the robust approximation properties of RBFs (Buhmann, 2000) on which RBF networks are based (Giroso & Poggio, 1990; Zemouri et al., 2003) and RBF network layers are added to existing RC-RNNs (Itoh & Adachi, 2010; Tyshchenko, 2016). For each Gaussian RBF projector $\mathcal{A} = \exp(-\gamma(y_1(x, t) - y_c(x))^2)$, there are two parameters: (i) a shape factor γ and (ii) a space-dependent centre y_c . As in RBF networks, we choose the centre in an unsupervised manner and optimize for γ via a trial-and-correction procedure. In the GS system, $y_1(x, t)$ varies from 0 to 1, and the mean strength of the nonlinear term increases as $y_1(x, t)$ varies from 1 to 0. We therefore choose $y_c(x) = 0$ as the centre parameter ($y_c(x) = 1$ is an equally good choice). Fig. 3 shows that the GSG_{rbf} network, which uses a Gaussian RBF projector $\exp(-2y_1^2)$, performs noticeably better, both in terms of short-term accuracy as well as the statistics of the predictions, as compared to any of the GSPⁿ networks. The results are further improved, particularly in terms of the statistics, when two Gaussian RBF projectors ($\exp(-2y_1^2)$ and $\exp(-3y_1^2)$) are used in the GS2G_{rbf} network.

Fig. 3 (c) shows the VPT from the GSP⁵ and GS2G_{rbf} networks, which highlights the significant improvement in the predictive accuracy when Gaussian RBFs are used instead of monomials as nonlinear projectors. The small VPT values from all the networks indicate the difficulty in obtaining accurate predictions from partial observations. This difficulty can be further appreciated from Fig. 6 in which the governing equations are used for obtaining the predictions.

5. Relevance of nonlinear approximation properties of nonlinear projectors

We explore the connection between the nonlinear approximation properties of the projector functions and the performance of the corresponding ESNs that are based on those projector

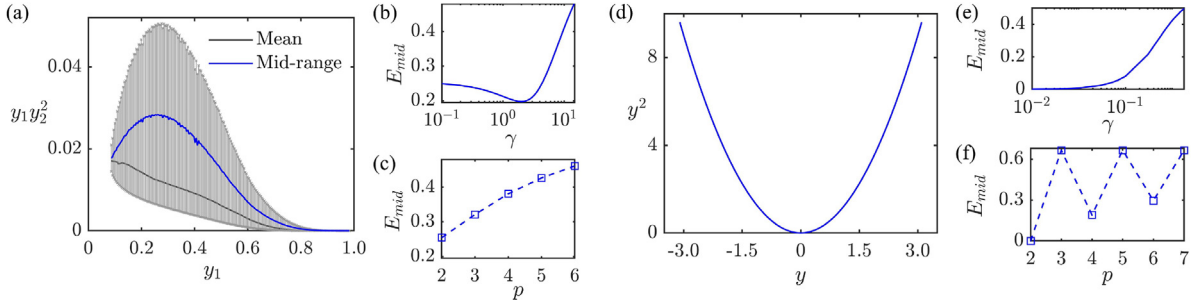


Fig. 4. Nonlinear terms (a, d) and errors in approximating their mid-range values via Gaussian RBFs (b, e) and monomials (c, f) for the GS system (a–c) and the KS system (d–f). For the GS system, the Gaussian RBFs have relatively lower error, thus explaining the superior performance of the networks with the Gaussian RBF projectors. For the KS system, even when the nonlinearity is a monomial, the Gaussian RBF can approximate the nonlinear term well.

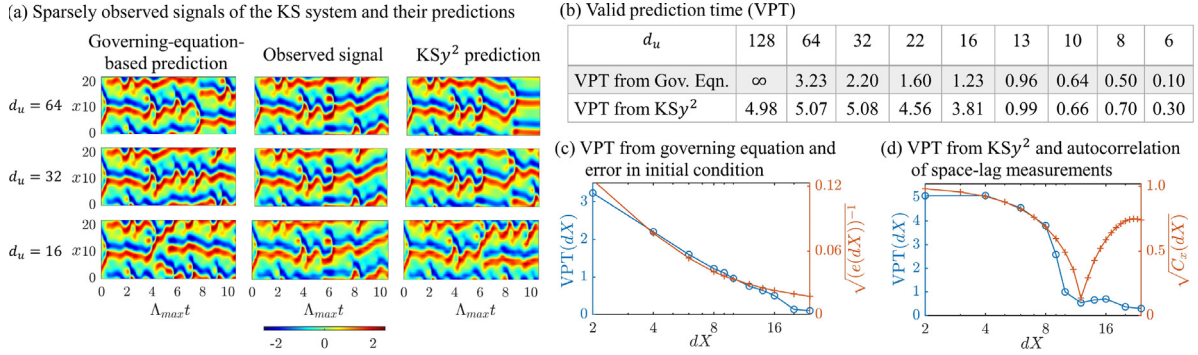


Fig. 5. (a) Sparsely observed signals of the KS system and their predictions from the governing equations and the KSy² network ($d_r = 2000$ and $N_g = 1$). (b) The VPT of predictions at different levels of sparsity. The observed signal becomes increasingly sparse as d_u decreases (or $dX \approx 128/d_u$ increases). (c) The VPT for the governing-equations-based predictions decreases rapidly with increasing dX , varying with the inverse of the square-root of the root-mean-square-error in the initial condition ($e(dX)$). (d) The VPT for the KSy² predictions are relatively unaffected as long as the autocorrelation between observations at neighbouring spatial locations ($C_x(dX)$) remains close to one.

functions. Fig. 4(a) shows the variation of the nonlinear term $y_1 y_2^2(y_1)$, its mean and mid-range values (m) with the observed variable y_1 in the GS system. The nonlinear term $y_1 y_2^2$ is calculated using y_1 and y_2 time-series data, the shaded region in the plot shows the range of values of the nonlinear term $y_1 y_2^2$, and m is the mean of the maximum and minimum values of $y_1 y_2^2$ at a given y_1 . The error in approximating m by a nonlinear projector function ($\mathcal{A}$) is given as

$$E_{mid} = \min_{a_1, a_2} \sqrt{\langle (a_1 + a_2 \mathcal{A} - m)^2 \rangle / \langle m^2 \rangle}, \quad (4)$$

where $\mathcal{A} = \exp(-\gamma y_1^2)$ for Gaussian RBFs (the corresponding E_{mid} is shown in subfigure (b)) and $\mathcal{A} = y_1^p$ for monomials (the corresponding E_{mid} is shown in subfigure (c)). The angular brackets denote spatial averaging. There are three points to note here. First, the value of γ at which GSG_{rbf} performs best coincides with that at which E_{mid} is minimum (i.e. $\gamma \approx 2$), demonstrating that a good nonlinear projector does indeed approximate the system nonlinearity. Second, minimization of the error with respect to the mid-range value (rather than the mean value) indicates that a good nonlinear projector minimizes the chances of maximum error (i.e. L_∞ -error) (Giloni & Padberg, 2002). Third, E_{mid} is lower for the Gaussian RBFs than for the monomials, thus explaining the superior performance of the GSG_{rbf} and $GS2G_{rbf}$ networks over the GSP^n networks in Fig. 3. Such statistical interpretation indicates that RBFs can be useful in other RNNs as well.

Furthermore, the robustness of RBFs is evidenced by their ability to satisfactorily approximate most types of nonlinearities. For example, $\mathcal{A} = y^2$ is the best projector for the KS system for which the nonlinear term itself is y^2 . However, Gaussian RBFs can still approximate y^2 nonlinearity well (see Fig. 4(e)) and the corresponding ESN (KSG_{rbf}) predicts the KS system relatively

well (see Fig. 2 (b–c)). Here we have only shown the results for Gaussian RBFs; the results are similarly good when multiquadric RBFs are used (see Appendix C).

6. Comparison between model-free and governing-equations-based predictions

6.1. Application to Kuramoto–Sivashinsky system with sparse measurements

We now consider a situation where only sparse measurements of the KS system are available. We compare the predictive performance of the governing-equations-based time-marching with that of the KSy² network. We take the ground truth as the original simulation with $N = 128$ gridpoints. The observed signal ($y_{obs}(x, t)$), shown in the middle column of Fig. 5 (a), consists of measurements at d_u equispaced points at distance dX/N . This implies that the observations become increasingly sparse with decreasing d_u (or increasing dX).

The governing-equations-based predictions are obtained by time-marching from the initial condition at $t = 0$. The initial condition, say $y_{int}(x)$, is obtained by linearly interpolating $y_{obs}(x, t = 0)$ from the observed space (consisting of d_u points) to the numerical grid (consisting of $N = 128$ points). The predictions and the VPT for different levels of sparsity in the observations are shown in Fig. 5 (a) and (b), respectively. The accuracy of the predictions decreases with increasing sparsity as indicated by the reduction in VPT. Fig. 5 (c) shows that the reduction in VPT is proportional to $\sqrt{e(dX)^{-1}}$, where $e(dX) = \sqrt{\langle (y(x, t = 0) - y_{int}(x, t = 0))^2 \rangle}$ is the root-mean-square error in the initial condition. We note that the error in the initial conditions could be reduced by using

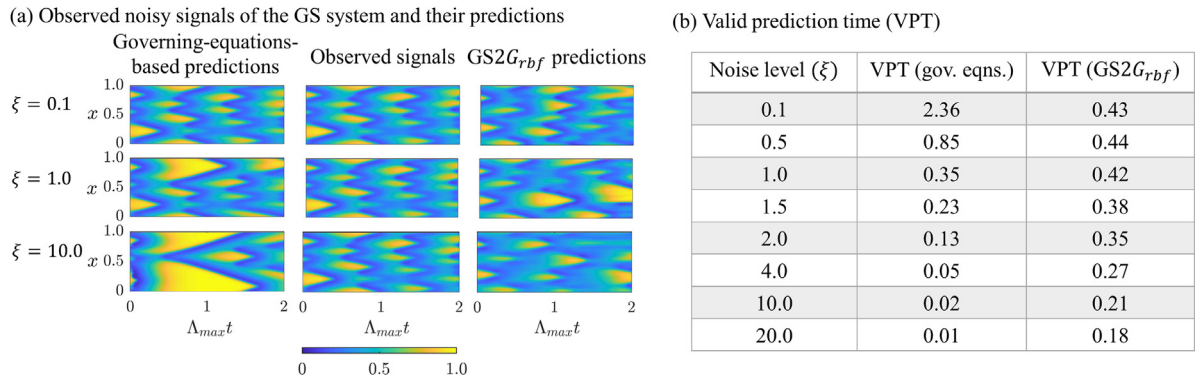


Fig. 6. (a) Partially observed noisy signals for the GS system and their predictions from the governing equations and the GS2Grbf network (of size $d_r = 1500$ and $N_g = 4$). (b) VPT for the predictions from the governing equations and the GS2Grbf network. A reduction in predictive accuracy occurs with increasing noise level. The GS2Grbf network, however, is relatively less affected by the noise. The small values of VPT from both methods demonstrate the challenge of forecasting chaotic systems with partial observability.

other interpolation schemes or data assimilation (Brajjard, Carrassi, Bocquet, & Bertino, 2020; Chandramouli, Memin, & Heitz, 2020; Wang & Zaki, 2021; Whitaker, Compo, & Thépaut, 2009). However, the prediction error would still increase as the information at smaller scales becomes unavailable. We also note that the system parameters and boundary conditions are known perfectly here, which often contribute to error in the governing-equations-based predictions.

The predictions from the KSy² network are obtained by training the network with the sparsely observed data, i.e. the network never sees the original data. The predictions and VPT from the KSy² network for different levels of sparsity in the observations are shown in Fig. 5 (a) and (b), respectively. We find that the predictions from the RC-RNN are insensitive to the discretization or interpolation error; a similar level of accuracy is maintained even when d_u is reduced from N to $N/4$. Fig. 5 (d) shows that VPT from KSy² is proportional to $\sqrt{C_x(dX)}$, where $C_x(dX)$ is the mean autocorrelation between $y(x)$ and $y(x + dX/N)$ signals. The results show that information about the well correlated elements of the state-vector can be learned from the time history of partial measurements. This suggests that RC-RNNs can unravel the space-time dynamics in agreement with Takens' embedding theorem (Takens, 1981), which has recently been suggested to be the case also for data-driven discovery of the governing equations from partial measurements (Bakarji, Champion, Kutz, & Brunton, 2022).

The results show that RC-RNNs can provide excellent predictions up to moderate levels of sparsity even when the governing equations may suffer from high discretization errors. The accuracy of both methods can be improved, such as by using better interpolation schemes for the governing equations or by increasing the reservoir dimension for the RC-RNN. The key point, however, is that RC-RNNs are less affected by the sparsity as compared to governing-equations-based time-marching. When the sparsity is high such that the spatially adjacent measurements are beyond the first zero in the autocorrelation, we find that the RC-RNNs perform poorly. We expect the same limitations for other data-driven methods as well.

6.2. Application to Gray-Scott system with partial and noisy measurements

We now consider another practical scenario in which only partial and noisy observables are available. We take the ground truth as the original simulation of the GS system. The concentration of species one with measurement noise, say y_{1obs} , is the observable (shown in the middle column of Fig. 6 (a)). The measurement noise is additive, white in space and time, with uniform

distribution between $[-\xi\sigma_{y_1}, \xi\sigma_{y_1}]$, where σ_{y_1} is the standard deviation of y_1 and ξ is the noise level. Similar to the sparse measurement cases, the governing-equations-based predictions are obtained by time-marching from the initial conditions that are calculated from the noisy measurements of y_1 at $t = 0$, while RC-RNN-based predictions are obtained by training the GS2Grbf network with the noisy measurements.

The governing equations, their parameters and the boundary conditions are all known perfectly, but the initial condition for y_2 is now estimated from the noisy measurements of y_1 . This introduces high levels of error in the initial condition and, consequently, the accuracy of the governing-equations-based predictions drops drastically with increasing noise-level as shown in Fig. 6. The RC-RNNs are fundamentally different in that they predict the future by learning patterns in the time history. By design, therefore, they are robust to small levels of measurement noise. The results in Fig. 6 show that although the accuracy of the RC-RNN-based predictions also decrease with increasing noise level, they do not suffer as much as the governing-equations-based predictions.

7. Conclusions

The main conclusions of this paper are threefold. The first two concern the explicitly added nonlinearity in RC-RNNs. Fig. 2 and Chattopadhyay et al. (2020), Pathak et al. (2018) show it to be essential to explicitly add nonlinearities in RC-RNNs to enable them to capture complex system nonlinearities. Firstly, we find it more computationally efficient to add the nonlinearity before the reservoir, such as in the KSy² network, than after the reservoir, such as in the KSr² network. Adding the nonlinearity before the reservoir preconditions the network towards the system dynamics and hence approximate the system nonlinearities better. Secondly, we find that the use of RBFs as nonlinear projectors (to explicitly add nonlinearities in RC-RNNs) enables the networks to capture the system nonlinearities even with partial measurements and/or without knowing the governing equations. We show that the robust approximation properties of RBFs, which makes them suitable for nonlinear approximation in numerical schemes and RBF networks (Buhmann, 2000; Girosi & Poggio, 1990; Zemouri et al., 2003), also makes them a good choice as nonlinear projectors in the RC-RNNs proposed here. The best nonlinear projectors approximate the mid-range value of the system nonlinearities, thus minimize the L_∞ -error. This statistical interpretation enhances the viability of using RBFs in other RNNs for turbulence prediction.

The third conclusion is on viability of RC-RNNs as an alternative to conventional methods in practical situations. Observations

in natural and engineering systems can often have missing data, coarse resolution and noise; thus several ML methods have been developed to extract information from limited and noisy measurements (Raissi, Yazdani, & Karniadakis, 2020; Reichstein et al., 2019). Prediction of turbulent systems, however, is extremely sensitive to any estimation errors. This is evident from the results in Figs. 5 and 6, which show that the governing equations, even with perfectly known boundary conditions and parameters, quickly lose predictive accuracy as the measurements become sparse, incomplete and noisy. We find that the predictive performance of RC-RNNs is relatively less affected by sparsity and noise in the measurements, thus can potentially outperform the governing equations in such cases.

In summary, we have demonstrated the capabilities of our proposed RC-RNN framework to forecast (i) partially observable spatiotemporally chaotic systems with no knowledge of the form of system nonlinearity and (ii) sparsely observable systems or partially observable systems with measurement noise for which even the governing equations become inaccurate. No recourse to the governing equations is required, thus making our RC-RNN framework truly model-free. We emphasize that the partial and sparse observability in our work is fundamentally different from the model-reduction or partial observability discussed in recent studies on turbulence forecasting (Vlachas et al., 2022, 2018). In those studies, the extraction of dominant system modes may require noise-free measurements of the full state-vector. Finally, we also highlight the difficulty in predicting noisy and partially observable systems, indicating that much larger reservoirs than those used in this study may be required for useful predictions in real-world applications, such as weather forecasting. Understanding the relationship between the system dynamics and hyperparameters of RC-RNNs and the uncertainty quantification of the predictions would be vital to unlocking the full potential of applying RC-RNNs to practical turbulent systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Details of the reservoir networks

In an echo-state network (ESN), $W_{in} \in \mathbb{R}^{d_r \times d_u}$ connects the input vector $\mathbf{u}(t) \in \mathbb{R}^{d_u}$ to the reservoir nodes where the elements of the input vector are scaled by a scalar σ . Following Pathak et al. (2018), each reservoir node is connected to a single element from the input vector, while each element of the input vector is connected to d_r/d_u reservoir nodes. Connecting each reservoir node by a single input element ensures that uncorrelated input elements, such as from two spatially uncorrelated locations, do not get mixed in the reservoir. The non-zero elements of W_{in} are randomly chosen from a normal distribution in $[-1, 1]$. In the networks with nonlinear projectors, such as the KSG_{rbf}, GSPⁿ, GSG_{rbf} and GS2G_{rbf} networks, the size of the input vector is increased from d_u to $(L+1)d_u$, where L is the number of nonlinear projectors. The elements of the input vector from each nonlinear projector are scaled independently by respective scalars σ_i , where $i = 1, 2, \dots, L$ corresponds to each nonlinear projector. We note that the size of the matrix W_{in} also increases from $d_r \times d_u$ to $d_r \times (L+1)d_u$. This has no significant impact on the computational cost or memory requirements in training the network, which mostly depend on the reservoir size d_r and the length of the training time-series (see Eq. (A.1)). However, networks with more nonlinear projectors have more hyperparameters to be optimized and hence it may take longer to optimize their performance.

The adjacency matrix $A \in \mathbb{R}^{d_r \times d_r}$ determines the internal connections between the reservoir nodes. The elements of A are also randomly chosen from a normal distribution in $[-1, 1]$, they are then rescaled such that the largest eigenvalue of A (also known as the spectral radius) is equal to ρ . It is sometimes recommended to make A sparse by randomly fixing most of its elements to be equal to 0. However, we do not find any advantage in doing that in our case.

The elements of $W_{out} \in \mathbb{R}^{d_v \times d_r}$ are optimized to minimize the squared-error between the output $\mathbf{v}(t)$ and the target signal $\hat{\mathbf{v}}(t)$. The optimized elements are obtained by performing the linear regression with Tikhonov regularization:

$$W_{out} = \hat{\mathbf{V}}\mathbf{R}^T (\mathbf{R}\mathbf{R}^T + \beta\mathbf{I})^{-1}, \quad (\text{A.1})$$

where $\hat{\mathbf{V}} = [\hat{\mathbf{v}}(-T + N_{tran}\Delta t), \hat{\mathbf{v}}(-T + N_{tran}\Delta t + \Delta t), \dots, \hat{\mathbf{v}}(-\Delta t)]$, $\mathbf{R} = [\mathbf{r}(-T + N_{tran}\Delta t), \mathbf{r}(-T + N_{tran}\Delta t + \Delta t), \dots, \mathbf{r}(-\Delta t)]$, $\beta = 1e^{-5}$ is the regularization parameter and $\mathbf{I}$ is an identity matrix. The vectors $\hat{\mathbf{v}}(-T)$, $\hat{\mathbf{v}}(-T + \Delta t)$, $\dots$, $\hat{\mathbf{v}}(-\Delta t)$ and $\mathbf{r}(-T)$, $\mathbf{r}(-T + \Delta t)$, $\dots$, $\mathbf{r}(-\Delta t)$ are the target signal and reservoir states during the training period, i.e. from $t = -T$ to 0 and Δt is the discrete time step of the network. Because the reservoir state at $t = -T$ is initialized to 0, $N_{tran} = 50$ number of initial transient steps (which is much smaller than the total training time steps) are discarded during the training.

For the implementation of the parallelized network, the local nature of interactions in spatiotemporally chaotic systems is exploited and the system is divided into spatially localized regions. Consider the input vector as $\mathbf{u}(t) = [u_1, u_2, \dots, u_{d_u}](t)$ and the output vector as $\mathbf{v}(t) = [v_1, v_2, \dots, v_{d_v}](t)$. Each reservoir receives $d_u/N_g + 2N_i$ elements of the input vector and predicts d_v/N_g elements of the output vector, such that the j th reservoir receives $[u_{jd_u/N_g+1-N_i}, \dots, u_{jd_u/N_g+1}, \dots, u_{(j+1)d_u/N_g}, \dots, u_{(j+1)d_u/N_g+N_i}]$ elements and predicts $[v_{jd_v/N_g+1}, \dots, v_{(j+1)d_v/N_g}]$ elements, where N_g is the number of reservoirs and N_i is the number of overlapping elements of the input vector between adjacent reservoirs (see Pathak et al., 2018 for details). For the modified networks with nonlinear projectors, each reservoir receives the same $d_u/N_g + 2N_i$ elements from the input vector as before and additionally also receives their nonlinear projections.

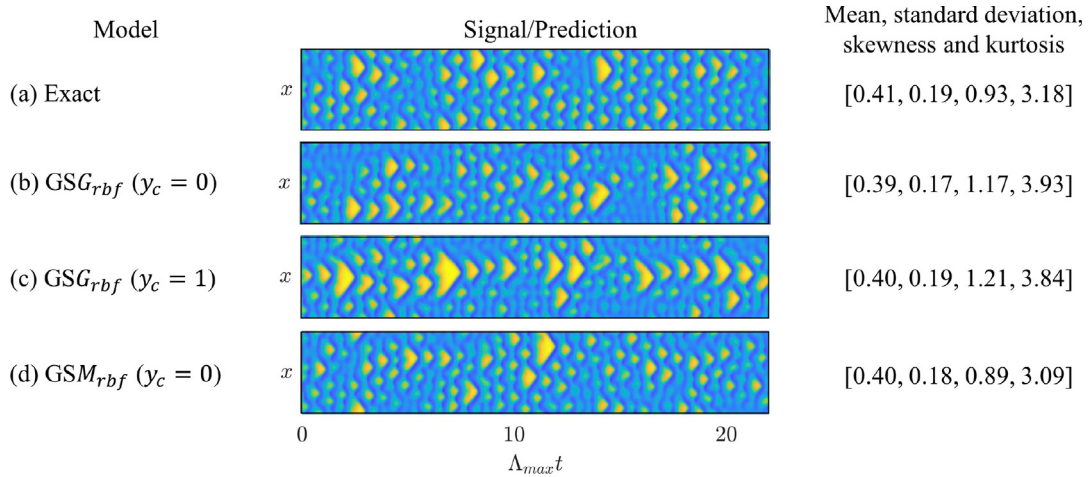


Fig. C.7. (a) The observed signal y_1 of the Gray–Scott system and (b–d) its predictions from different RC-RNNs. The networks $GSG_{rbf} (y_c = 0)$ uses $\exp(-2y_1^2)$, $GSG_{rbf} (y_c = 1)$ uses $\exp(-2(y_1 - 1)^2)$ and $GSM_{rbf} (y_c = 0)$ uses $\sqrt{1 + 3y_1^2}$ as nonlinear projectors.

Appendix B. Calculations of the mean prediction error and the VPT

There are two sources of randomness in the calculations of the prediction error ($\mathcal{E}(t)$): (i) the initial condition of the system and (ii) the random seed used when drawing the elements of W_{in} and A matrices. Once the network is trained using data from $t = -T$ to 0 , we calculate the predictions by reinitializing the network for 200 initial conditions. Given that ESNs satisfy the fading memory property, we only need a finite sequence of past inputs, say from $t_r - N_{Relnit} \Delta t$ to $t_r - \Delta t$, to obtain the predictions starting from t_r . We find that a finite sequence of $N_{Relnit} = 10$ past inputs is sufficient to reinitialize the ESNs in our study. During this reinitialization, the activation state is set to zero, i.e. $\mathbf{r}(t_r - N_{Relnit} \Delta t) = 0$, and is then evolved using the inputs $\mathbf{u}(t - N_{Relnit} \Delta t)$, $\mathbf{u}(t_r - N_{Relnit} \Delta t + \Delta t)$, ..., $\mathbf{u}(t_r - \Delta t)$. Such calculations are repeated for 10 random seeds. The mean prediction error ($\bar{\mathcal{E}}(t)$) is obtained by averaging the prediction error ($\mathcal{E}(t)$) over 200 initial conditions and 10 random seeds, i.e. for 2000 calculations. The VPT is then calculated as the nondimensional time at which $\bar{\mathcal{E}}(t)$ crosses the threshold 0.5.

The mean prediction error ($\bar{\mathcal{E}}(t)$) presented in the paper is approximately optimized over all the hyperparameters for each network. The main hyperparameters in a network of given size d_r and the number of parallel units N_g are: (i) the interaction length N_i , (ii) the spectral radius ρ of A and (iii) the weights (σ , σ_i) of input elements. When nonlinear projector functions are used, there are additional hyperparameters associated with the projector functions.

Appendix C. Different radial basis functions

In the main text, we only used Gaussian radial basis functions (RBFs) as nonlinear projector for which we fixed the centre at $y_c(x) = 0$ to predict the Gray–Scott system for which only concentration of one species, $y_1(x, t)$, is observable. Fig. C.7 shows that using Gaussian RBF with centre at $y_c = 1$ and multiquadric RBF are equally good options, thus showing the robustness of RBFs as nonlinear projectors. A multiquadric RBF is given as $\sqrt{1 + \gamma(y_1 - y_c)^2}$, where γ is the shape factor and y_c is the centre.

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